

Eduardo Vasconcelos

Security Engineer | Systems Programming | Android | Linux Kernel | MSc | PhD Student

Contact

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Key Highlights

- Close to 10 years of experience in Security Engineering with emphasis on Android;
- Active contributor to the Linux kernel with 3 patches contributed to upstream;
- MSc in Computer Science awarded for national-level excellence in Software Engineering research in 2024;
- Computer Engineer awarded for excellence in the undergraduate thesis in 2016;
- Experience in teaching Python programming in vocational/technical education;
- Proficient in three languages: Portuguese (*native*), English (*IELTS: C1*), and French (*TCF: B2-C1*).

Skills Matrix

Level	Skill
Advanced	Threat Modeling, Security Testing, Security Code Review, Research & Development
Intermediate	Reverse Engineering, DevSecOps, Cryptography, Linux SysAdmin, Linux Kernel Development
Familiar	Incident Response, Embedded Systems

Work History

- **Master Application Security Architect** @ SiDi, Brazil (remote)

Aug 2024–Present

- Reverse engineered Android malware, decompiling DEX to Java and analyzing x86 shared objects to develop malicious behavior heuristics and Machine Learning models for malware detection;
- Designed and implemented automated Android malware analysis tools in Python to unveil and monitor malicious application behavior;
- Performed threat modeling and security testing of Android and web applications, including Samsung Knox;
- Participated in internal AI red teaming task forces, contributing to AI security testing knowledge building and reporting security vulnerabilities in AI agents;
- Key contributor in revitalizing the Security Champions program.

- **Open Source Contributor** @ Linux kernel

Apr 2026–Present

- Contributed a total of 3 patches to the upstream Linux kernel;
- Currently focusing on the Linux Security Module (LSM) framework and AppArmor.

- **Technical Instructor** @ SENAI, Brazil (on-site)

May 2025–Jan 2026

- Taught Python programming classes.

- **Senior Mobile Security Engineer** @ iFood, Brazil (remote)

Feb 2022–Aug 2024

- Designed and implemented security and fraud prevention mechanisms in C++, integrated into mission-critical Android applications via Java Native Interface (JNI), serving over 40 million monthly users;
- Reverse engineered malicious Android apps, decompiling DEX to Java and analyzing x86 shared objects, to reveal methods and behaviors used by fraudsters in food delivery services;
- Integrated Runtime Application Self-Protection (RASP) solutions into GitLab CI/CD pipelines for mobile apps.

- **Senior Software Security Analyst** @ SiDi, Brazil

Apr 2019–Feb 2022

- Delivered ongoing offensive security testing services to Samsung, emphasizing Android applications and Android OS system services;

- Conducted research on Android API and OS vulnerabilities on behalf of Samsung, including issues related to Android accessibility services abuses and the permissions system circumvention.
- **Information Security Analyst @ Hacker Rangers, Brazil**
Jun 2017–Apr 2019
 - Delivered consulting services in web application and network security assessments and phishing campaigns;
 - Developed and operated the Hacker Rangers SaaS security awareness training platform;
 - Administered IBM Security Guardium virtual appliances built on Red Hat Linux, handling deployment, configuration, and operation for private organizations and government agencies.
- **Information Technology Intern @ Embraer, Brazil**
Sep 2016–Jun 2017
 - Contributed to the development of an IT Change Management plan;
 - Developed Python tools to analyze corporate web traffic.
- **Engineering Intern @ SigMEDIA, Ireland**
May 2014–Jul 2014
 - Developed a Machine Learning ensemble model to detect laughter in human speech;
 - Processed digital voice signals in Python.

Education

- **Doctoral Degree in Computer Science @ USP, Brazil**
Apr 2026–Present
 - Research topic: fuzz testing embedded systems operating in brownout voltage regimes for security bugs;
 - Currently serving as a volunteer Debian Linux system administrator at the Software Engineering Lab.
- **Master’s Degree in Computer Science @ USP, Brazil**
Feb 2022–Aug 2025
 - Graduated with an A average;
 - Research topic: leveraging Mutation Testing techniques to inject vulnerabilities into Android apps, providing a novel approach to evaluating mobile security testing procedures;
 - Received the Distinguished Paper Award at the Brazilian Conference on Software 2024 (CBSOFT’24) for my work, “Mutation Testing to Support the Security Testing of Android Applications”;

- Served as a teaching assistant for Software Engineering.
- **Postgraduate Diploma in Software Engineering @ Unicamp, Brazil**
Feb 2019–Dec 2019
 - Graduated with a GPA of 9.4 out of 10.0;
 - One-year Postgraduate Diploma program in Software Engineering.
- **Engineer’s Degree in Computer Engineering @ USP, Brazil**
Feb 2012–Jul 2017
 - Graduated 13th out of 59 with a GPA of 7.5 out of 10.0;
 - Awarded the Sao Paulo Research Foundation (FAPESP) undergraduate research scholarship, contributing to a research project focused on flood forecasting in urban areas;
 - Received the Best Computer Engineering Undergraduate Thesis Award in 2016 for my thesis on emotion recognition in human speech using Machine Learning;
 - Served as a teaching assistant for Python programming;
 - Served as a teaching assistant for Java and C++ programming;
 - Served as a volunteer PHP developer at the junior enterprise.
- **Visiting Student in Electronic Engineering @ TCD, Ireland**
Sep 2013–May 2014
 - Full-term visiting student, earning 60 ECTS with honors;
 - Awarded the Brazilian Federal Government’s “Science without Borders” (SwB) scholarship;
 - Received a Special Prize for the Best Product Branding Video at the Trinity Engineering Student Entrepreneurs of the Year Competition in 2014.
- **Professional Training in Computer Hardware and Network Maintenance @ Microlins, Brazil**
Mar 2007–Sep 2007
 - Six-month professional training program in computer hardware, operating systems, and computer networks.